WORKED SOLUTION

STEP III 2020, Question 4

Pure • Reflection and Rotation Matrices

STEP III 2020, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

The opening result is the whole question in disguise. Subtracting $(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$ takes a point straight down to the plane, so subtracting *twice* that amount carries it the same distance out the other side, which is a reflection. Once a reflection is written as $\mathbf{x}\mapsto\mathbf{x}-2(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$, its matrix, a half-turn about $\mathbf{n}$, and their product all follow with very little calculation.

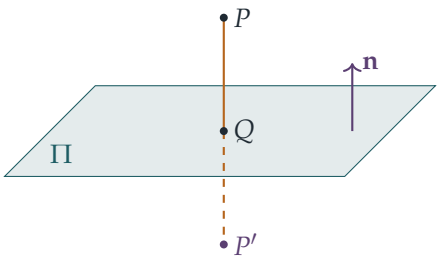
Solution

The opening result

Let $\mathbf{q}=\mathbf{x}-(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$. Since $\mathbf{n}\cdot\mathbf{n}=1$,

$$\mathbf{q}\cdot\mathbf{n}=\mathbf{x}\cdot\mathbf{n}-(\mathbf{x}\cdot\mathbf{n})(\mathbf{n}\cdot\mathbf{n})=0,$$

so Q lies on Π. Also $\overrightarrow{PQ}=\mathbf{q}-\mathbf{x}=-(\mathbf{x}\cdot\mathbf{n})\mathbf{n}$, which is a multiple of the normal $\mathbf{n}$. It is non-zero because P is not on Π (so $\mathbf{x}\cdot\mathbf{n}\neq 0$). Hence PQ is perpendicular to Π, as required.



The picture also shows where we are heading: the mirror image P' of P in Π is as far beyond Q as P is above it, so its position vector is

$$\mathbf{x}-2(\mathbf{x}\cdot\mathbf{n})\mathbf{n}.$$

(If P is on Π this formula gives P itself, which is also correct.)

Part (i)

Take $\mathbf{n} = (a, b, c)^{\mathsf{T}}$, a unit normal because $a^2 + b^2 + c^2 = 1$. For $\mathbf{i}$ we have $\mathbf{i} \cdot \mathbf{n} = a$, so

$$T(\mathbf{i}) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 2a \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 - 2a^2 \\ -2ab \\ -2ac \end{pmatrix} = \begin{pmatrix} b^2 + c^2 - a^2 \\ -2ab \\ -2ac \end{pmatrix},$$

using $1 = a^2 + b^2 + c^2$ in the top entry. This is the required image. In the same way, with $\mathbf{j} \cdot \mathbf{n} = b$ and $\mathbf{k} \cdot \mathbf{n} = c$,

$$T(\mathbf{j}) = \begin{pmatrix} -2ab \\ a^2 + c^2 - b^2 \\ -2bc \end{pmatrix}, \qquad T(\mathbf{k}) = \begin{pmatrix} -2ac \\ -2bc \\ a^2 + b^2 - c^2 \end{pmatrix}.$$

The columns of $\mathbf{M}$ are the images of $\mathbf{i}, \mathbf{j}, \mathbf{k}$, so

$$\mathbf{M} = \begin{pmatrix} b^2 + c^2 - a^2 & -2ab & -2ac \\ -2ab & a^2 + c^2 - b^2 & -2bc \\ -2ac & -2bc & a^2 + b^2 - c^2 \end{pmatrix} = \mathbf{I} - 2\mathbf{n}\mathbf{n}^{\mathsf{T}}.$$

Part (ii)

We match the given matrix with $\mathbf{M}$, writing each diagonal entry as $1 - 2a^2$, $1 - 2b^2$, $1 - 2c^2$:

$$1 - 2a^2 = 0.64, \quad 1 - 2b^2 = 0.36, \quad 1 - 2c^2 = 0 \implies a^2 = 0.18, \quad b^2 = 0.32, \quad c^2 = 0.5.$$

So $|a| = \frac{3}{5\sqrt{2}}$, $|b| = \frac{4}{5\sqrt{2}}$ and $|c| = \frac{5}{5\sqrt{2}}$.

? **Ask yourself** The diagonal only gives a^2, b^2, c^2. How do we settle the signs?

The off-diagonal entries give the products:

$$-2ab = 0.48, \quad -2ac = 0.6, \quad -2bc = -0.8 \implies ab = -0.24, \quad ac = -0.3, \quad bc = 0.4.$$

So b and c have the same sign, and a has the opposite sign. Replacing (a, b, c) by $(-a, -b, -c)$ changes nothing (it is the same plane), so we may take $c > 0$, which gives

$$(a, b, c) = \tfrac{1}{5\sqrt{2}}(-3,\ 4,\ 5).$$

These values give $ab = -\frac{12}{50}$, $ac = -\frac{15}{50}$ and $bc = \frac{20}{50}$, so all six independent entries match. Multiplying through by $-5\sqrt{2}$, the plane is

$$3x - 4y - 5z = 0.$$

Part (iii)

Split any vector as $\mathbf{x} = (\mathbf{x} \cdot \mathbf{n})\mathbf{n} + (\mathbf{x} - (\mathbf{x} \cdot \mathbf{n})\mathbf{n})$, a part along the axis plus a part perpendicular to it (the opening result). A rotation through π about the axis leaves the

first part alone and reverses the second, so

$$\mathbf{x} \mapsto (\mathbf{x}\cdot\mathbf{n})\mathbf{n} - (\mathbf{x} - (\mathbf{x}\cdot\mathbf{n})\mathbf{n}) = 2(\mathbf{x}\cdot\mathbf{n})\mathbf{n} - \mathbf{x}.$$

This is exactly $-T(\mathbf{x})$. Therefore $\mathbf{N} = 2\mathbf{n}\mathbf{n}^{\mathsf{T}} - \mathbf{I} = -\mathbf{M}$, that is,

$$\mathbf{N} = \begin{pmatrix} a^2 - b^2 - c^2 & 2ab & 2ac \\ 2ab & b^2 - a^2 - c^2 & 2bc \\ 2ac & 2bc & c^2 - a^2 - b^2 \end{pmatrix}.$$

Part (iv)

? **Ask yourself** Do we need to multiply two 3×3 matrices of a, b, c here?

No. From (iii), $\mathbf{NM} = -\mathbf{M}^2$. Reflecting twice in the same plane brings every point back to where it started, so $\mathbf{M}^2 = \mathbf{I}$. (Algebraically, $(\mathbf{I} - 2\mathbf{n}\mathbf{n}^{\mathsf{T}})^2 = \mathbf{I} - 4\mathbf{n}\mathbf{n}^{\mathsf{T}} + 4\mathbf{n}(\mathbf{n}^{\mathsf{T}}\mathbf{n})\mathbf{n}^{\mathsf{T}} = \mathbf{I}$, because $\mathbf{n}^{\mathsf{T}}\mathbf{n} = 1$.) Hence

$$\mathbf{NM} = -\mathbf{I},$$

which sends every $\mathbf{x}$ to $-\mathbf{x}$. This is reflection in the origin, also called central inversion or an enlargement with scale factor -1 centred at the origin. Notice that it doesn't depend on a, b or c at all.

Checks

- Each column of $\mathbf{M}$ should be a unit vector. For the first, $(b^2 + c^2 - a^2)^2 + 4a^2b^2 + 4a^2c^2 = (b^2 + c^2 + a^2)^2 = 1$.
- In (ii), the normal $(3, -4, -5)$ should be sent to its negative. The given matrix maps it to $(1.92 - 1.92 - 3,\ 1.44 - 1.44 + 4,\ 1.8 + 3.2) = (-3, 4, 5)$. Good.
- Taking $\mathbf{n} = \mathbf{k}$ makes $\mathbf{M} = \operatorname{diag}(1, 1, -1)$, reflection in $z = 0$, and $\mathbf{N} = \operatorname{diag}(-1, -1, 1)$, the half-turn about the z-axis. Their product is clearly $-\mathbf{I}$.
- A computer check with 1000 random unit vectors $\mathbf{n}$ confirmed the formula for $\mathbf{M}$, that $\mathbf{N}$ fixes $\mathbf{n}$ and reverses perpendicular vectors, and that $\mathbf{NM} = -\mathbf{I}$. The matrix in (ii) was reproduced to within 10^{-15}.

Common slips

- Using $\mathbf{x} - (\mathbf{x}\cdot\mathbf{n})\mathbf{n}$ as the reflection. That is the foot of the perpendicular, so the factor 2 is missing.
- In (ii), choosing the signs of a, b, c independently from the square roots. Only the off-diagonal entries fix the relative signs.
- Saying $\mathbf{NM}$ is "the identity". A reflection has determinant -1 and a rotation has determinant $+1$, so the product has determinant -1. In three dimensions $\det(-\mathbf{I}) = -1$, which is consistent.

Beyond STEP

The matrix $\mathbf{I} - 2\mathbf{n}\mathbf{n}^{\mathsf{T}}$ is a *Householder reflection*. Numerical linear algebra uses it constantly to reduce matrices to triangular form, and every orthogonal matrix in n dimensions is a product of at most n such reflections. Part (iv) is a small instance of that structure: in $\mathbb{R}^3$, a half-turn composed with the reflection in the perpendicular plane gives $-\mathbf{I}$. For orthogonal matrices, projections and reflections, see Strang, *Introduction to Linear Algebra*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.